Dylan Francis DAAN 682
15MAR2026

Homework_8:

Code:

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# DAAN: 682: Data Analytics Programming in Python
# Author: Dylan Francis
# Title: Homework_8: Supervised Learning with Scikit-learn: Classification Models
#
==============================================================================
#
# Quantitative Attributes
# Age   (years)
# BMI   (kg/m2)
# Glucose       (mg/dL)
# Insulin       (µU/mL)
# HOMA
# Leptin        (ng/mL)
# Adiponectin  (µg/mL)
# Resistin      (ng/mL)
# MCP-1(pg/dL)        (ng/mL)
# Labels
# 1 = Healthy Controls
# 2 = Patients

# Homework Questions:
#1.) Perform data exploratory data analysis (EDA) and interpret the results: display the data's
basic information and summary statistics, perform data quality checks, examine the distribution
of the target variable, and visualize the feature distributions. Note: A comprehensive EDA
should provide the essential foundation for data-driven decision-making when building
predictive models later. (10 points)
#2.) Use 30% of the data as the test set and build a Logistic regression model to predict the
Classification variable. (20 points)
#3.) Build a Naïve Bayes model to predict the Classification variable. (20 points)
#4.) Build a Decision tree model to predict the Classification variable. (20 points)
#5.) Build a Neural network model to predict the Classification variable. (20 points)
#6.) Which model is the best? Which variable is the most important one? (10 points)
#
==============================================================================

import os
import pandas as pd
import numpy as np
```

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from sklearn import datasets
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
from sklearn.tree import DecisionTreeClassifier
from sklearn.neural_network import MLPClassifier


path = r"C:\Users\dylan\OneDrive\Documents\GRAD_SCHOOL\DAAN_682\HOMEWORK_8"
target = "Classification"
os.chdir(path)
breastcancer = pd.read_csv("breastcancer.csv")

#1.) Data exploration
print(f"\nThere are {breastcancer.shape[0]} rows, and {breastcancer.shape[1]} columns.")
print("\n=======DATASET_INFO======")
print(breastcancer.info())
print("\n=======SUMMARY_STATS======")
print(breastcancer.describe())
breastcancer_corr = breastcancer.corr()
print(f"\n Here is the correlation martrix: {breastcancer_corr}")
print("\n===== MISSING VALUES =====")
print(f"{breastcancer.isnull().sum()}")
print("\n===== DUPLICATE ROWS =====")
print(f"The total number of duplicate rows is: {breastcancer.duplicated().sum()}")

# Here is the plot for the target variable (classification)
plt.figure(figsize=(6,4))
counter = sns.countplot(x=breastcancer[target])
for container in counter.containers:
    counter.bar_label(container)
plt.title("Target Variable Distribution")
plt.show()

# Here is the visualization of the feature variables
for col in breastcancer.columns:
    if(col != target):
        plt.figure(figsize=(6,4))
        sns.histplot(breastcancer[col], kde=True)
        plt.title(f"Feature Variable, {col} Distribution")
        plt.xlabel(col)
```

```
    plt.ylabel("Count")
    plt.show()

#2.)
features = breastcancer.drop(columns=[target])
targets = breastcancer[target]

features_train, features_test, targets_train, targets_test = train_test_split(
  features, targets, test_size=0.30, random_state=42
)

print(f"\nTraining set size: {features_train.shape}")
print(f"Test set size: {features_test.shape}")

log_model = LogisticRegression(max_iter=1000)
log_model.fit(features_train,targets_train)
targets_pred_log = log_model.predict(features_test)


print("\n MODEL PERFORMANCE for logistic Regression")
print(f"Accuracy: {accuracy_score(targets_test, targets_pred_log)}")

print("\n CONFUSION MATRIX FOR LOGISTIC REGRESSION")
print(confusion_matrix(targets_test, targets_pred_log))

print("\n CLASSIFICATION REPORT FOR LOGISTIC REGRESSION")
print(classification_report(targets_test, targets_pred_log))

#3.)
nb_model = GaussianNB()

nb_model.fit(features_train, targets_train)

targets_pred_nb = nb_model.predict(features_test)


print("\n NAIVE BAYES MODEL PERFORMANCE")
print(f"Accuracy: {accuracy_score(targets_test, targets_pred_nb)}")

print("\n CONFUSION MATRIX FOR BAYES NAIVE")
print(confusion_matrix(targets_test, targets_pred_nb))

print("\n CLASSIFICATION REPORT BAYES NAIVE")
print(classification_report(targets_test, targets_pred_nb))
```

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#4.)
dt_model = DecisionTreeClassifier(random_state=9)
dt_model.fit(features_train, targets_train)
targets_pred_dt = dt_model.predict(features_test)

print("\n DECISION TREE MODEL PERFORMANCE")
print(f"Accuracy: {accuracy_score(targets_test, targets_pred_dt)}")

print("\n CONFUSION MATRIX FOR DECISION TREE")
print(confusion_matrix(targets_test, targets_pred_dt))

print("\n CLASSIFICATION REPORT DECISION TREE")
print(classification_report(targets_test, targets_pred_dt))

from sklearn.tree import plot_tree

plt.figure(figsize=(20,10))
plot_tree(
    dt_model,
    feature_names=features.columns,
    class_names=[str(c) for c in dt_model.classes_],
    filled=True
)
plt.title("Decision Tree Visualization")
plt.show()

#5.)
nn_model = MLPClassifier(hidden_layer_sizes=(10,10),
                         activation = 'relu',
                         solver='adam',
                         max_iter=1000,
                         random_state=42)
nn_model.fit(features_train, targets_train)
targets_pred_nn = nn_model.predict(features_test)

print("\n NEURAL NETWORK MODEL PERFORMANCE")
print(f"Accuracy: {accuracy_score(targets_test, targets_pred_nn)}")

print("\n CONFUSION MATRIX for NEURAL NETWORK")
print(confusion_matrix(targets_test, targets_pred_nn))

print("\n CLASSIFICATION REPORT FORE NEURAL NETWORK")
print(classification_report(targets_test, targets_pred_nn))
```

6.)

```
print("\n MODEL PERFORMANCE for logistic Regression")
print(f"Accuracy: {accuracy_score(targets_test, targets_pred_log)}")
print("\n NAIVE BAYES MODEL PERFORMANCE")
print(f"Accuracy: {accuracy_score(targets_test, targets_pred_nb)}")
print("\n DECISION TREE MODEL PERFORMANCE")
print(f"Accuracy: {accuracy_score(targets_test, targets_pred_dt)}")
print("\n NEURAL NETWORK MODEL PERFORMANCE")
print(f"Accuracy: {accuracy_score(targets_test, targets_pred_nn)}")

print("\n CLASSIFICATION REPORT FOR LOGISTIC REGRESSION")
print(classification_report(targets_test, targets_pred_log))
print("\n CLASSIFICATION REPORT BAYES NAIVE")
print(classification_report(targets_test, targets_pred_nb))
print("\n CLASSIFICATION REPORT DECISION TREE")
print(classification_report(targets_test, targets_pred_dt))
print("\n CLASSIFICATION REPORT FORE NEURAL NETWORK")
print(classification_report(targets_test, targets_pred_nn))

print("Based on the accuracy, and on the classification reports, the neural network model
produce the best results.\n")
print("This model has the highest accuracy at: 0.8. It also produces the highest F1 score which is
a balance between precision and recall. ")
print("Based on the top node in the decision tree, we can see the most important variable is
glucose")
```

Terminal:

```
%runfile C:/Users/dylan/.spyder-py3/autosave/Homework_8.py --wdir

There are 116 rows, and 10 columns.

=======DATASET_INFO======
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 116 entries, 0 to 115
Data columns (total 10 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   Age           116 non-null    int64
 1   BMI           116 non-null    float64
 2   Glucose       116 non-null    int64
 3   Insulin       116 non-null    float64
```

```
 4   HOMA           116 non-null    float64
 5   Leptin         116 non-null    float64
 6   Adiponectin    116 non-null    float64
 7   Resistin       116 non-null    float64
 8   MCP.1          116 non-null    float64
 9   Classification 116 non-null    int64
dtypes: float64(7), int64(3)
memory usage: 9.2 KB
None

=======SUMMARY_STATS======
            Age        BMI  ...       MCP.1  Classification
count  116.000000  116.000000  ...   116.000000      116.000000
mean    57.301724   27.582111  ...   534.647000        1.551724
std     16.112766    5.020136  ...   345.912663        0.499475
min     24.000000   18.370000  ...    45.843000        1.000000
25%     45.000000   22.973205  ...   269.978250        1.000000
50%     56.000000   27.662416  ...   471.322500        2.000000
75%     71.000000   31.241442  ...   700.085000        2.000000
max     89.000000   38.578759  ...  1698.440000        2.000000

[8 rows x 10 columns]

 Here is the correlation martrix:                    Age       BMI  ...     MCP.1  Classification
Age            1.000000  0.008530  ...  0.013462       -0.043555
BMI            0.008530  1.000000  ...  0.224038       -0.132586
Glucose        0.230106  0.138845  ...  0.264879        0.384315
Insulin        0.032495  0.145295  ...  0.174356        0.276804
HOMA           0.127033  0.114480  ...  0.259529        0.284012
Leptin         0.102626  0.569593  ...  0.014009       -0.001078
Adiponectin   -0.219813 -0.302735  ... -0.200694       -0.019490
Resistin       0.002742  0.195350  ...  0.366474        0.227310
MCP.1          0.013462  0.224038  ...  1.000000        0.091381
Classification -0.043555 -0.132586  ...  0.091381        1.000000

[10 rows x 10 columns]

===== MISSING VALUES =====
Age            0
BMI            0
Glucose        0
Insulin        0
HOMA           0
Leptin         0
```

```
Adiponectin       0
Resistin          0
MCP.1             0
Classification    0
dtype: int64

===== DUPLICATE ROWS =====
The total number of duplicate rows is: 0

Training set size: (81, 9)
Test set size: (35, 9)

 MODEL PERFORMANCE for logistic Regression
Accuracy: 0.7428571428571429

 CONFUSION MATRIX FOR LOGISTIC REGRESSION
[[14  3]
 [ 6 12]]

 CLASSIFICATION REPORT FOR LOGISTIC REGRESSION
              precision    recall  f1-score   support

           1       0.70      0.82      0.76        17
           2       0.80      0.67      0.73        18

    accuracy                           0.74        35
   macro avg       0.75      0.75      0.74        35
weighted avg       0.75      0.74      0.74        35


 NAIVE BAYES MODEL PERFORMANCE
Accuracy: 0.6571428571428571

 CONFUSION MATRIX FOR BAYES NAIVE
[[15  2]
 [10  8]]

 CLASSIFICATION REPORT BAYES NAIVE
              precision    recall  f1-score   support

           1       0.60      0.88      0.71        17
           2       0.80      0.44      0.57        18

    accuracy                           0.66        35
```

```
   macro avg       0.70      0.66      0.64        35
weighted avg       0.70      0.66      0.64        35


 DECISION TREE MODEL PERFORMANCE
Accuracy: 0.7714285714285715

 CONFUSION MATRIX FOR DECISION TREE
[[12  5]
 [ 3 15]]

 CLASSIFICATION REPORT DECISION TREE
              precision    recall  f1-score   support

           1       0.80      0.71      0.75        17
           2       0.75      0.83      0.79        18

    accuracy                           0.77        35
   macro avg       0.78      0.77      0.77        35
weighted avg       0.77      0.77      0.77        35


 NEURAL NETWORK MODEL PERFORMANCE
Accuracy: 0.8

 CONFUSION MATRIX for NEURAL NETWORK
[[14  3]
 [ 4 14]]

 CLASSIFICATION REPORT FORE NEURAL NETWORK
              precision    recall  f1-score   support

           1       0.78      0.82      0.80        17
           2       0.82      0.78      0.80        18

    accuracy                           0.80        35
   macro avg       0.80      0.80      0.80        35
weighted avg       0.80      0.80      0.80        35


 MODEL PERFORMANCE for logistic Regression
Accuracy: 0.7428571428571429

 NAIVE BAYES MODEL PERFORMANCE
```

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Accuracy: 0.6571428571428571

DECISION TREE MODEL PERFORMANCE
Accuracy: 0.7714285714285715

NEURAL NETWORK MODEL PERFORMANCE
Accuracy: 0.8

CLASSIFICATION REPORT FOR LOGISTIC REGRESSION

	precision	recall	f1-score	support
1	0.70	0.82	0.76	17
2	0.80	0.67	0.73	18
accuracy			0.74	35
macro avg	0.75	0.75	0.74	35
weighted avg	0.75	0.74	0.74	35

CLASSIFICATION REPORT BAYES NAIVE

	precision	recall	f1-score	support
1	0.60	0.88	0.71	17
2	0.80	0.44	0.57	18
accuracy			0.66	35
macro avg	0.70	0.66	0.64	35
weighted avg	0.70	0.66	0.64	35

CLASSIFICATION REPORT DECISION TREE

	precision	recall	f1-score	support
1	0.80	0.71	0.75	17
2	0.75	0.83	0.79	18
accuracy			0.77	35
macro avg	0.78	0.77	0.77	35
weighted avg	0.77	0.77	0.77	35

CLASSIFICATION REPORT FORE NEURAL NETWORK

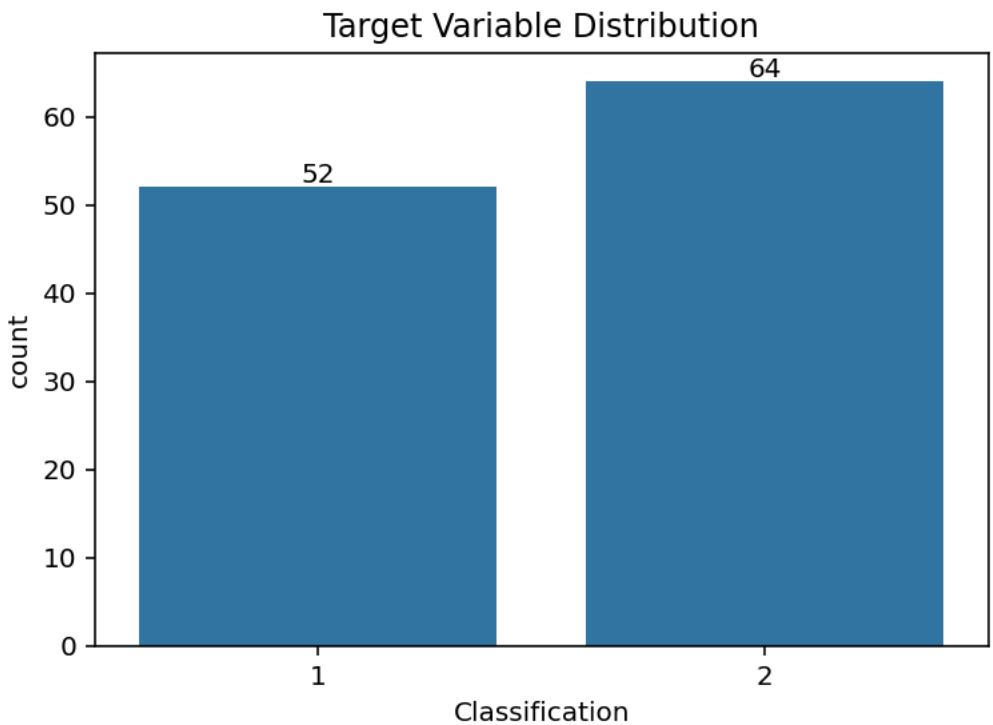
	precision	recall	f1-score	support

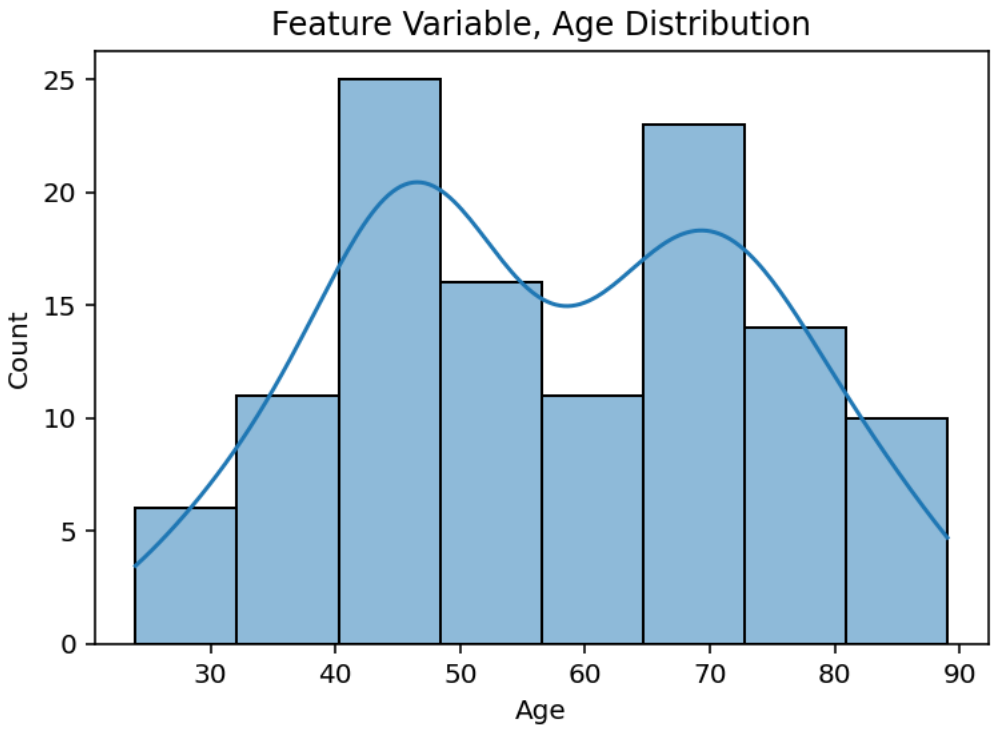
1	0.78	0.82	0.80	17
2	0.82	0.78	0.80	18
accuracy			0.80	35
macro avg	0.80	0.80	0.80	35
weighted avg	0.80	0.80	0.80	35

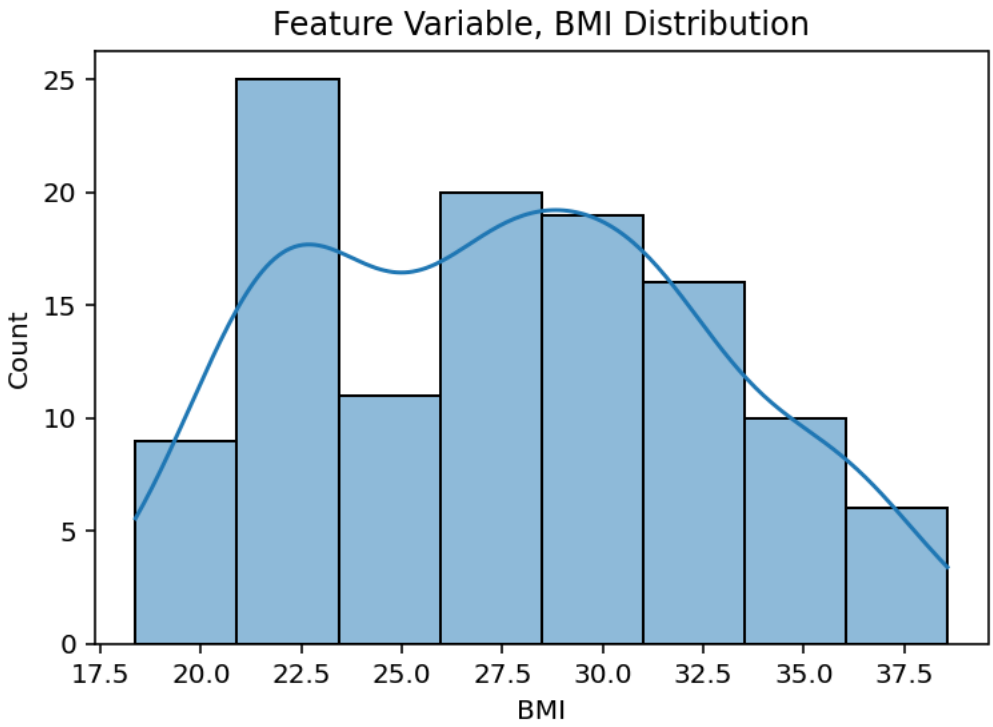
Based on the accuracy, and on the classification reports, the neural network model produce the best results.

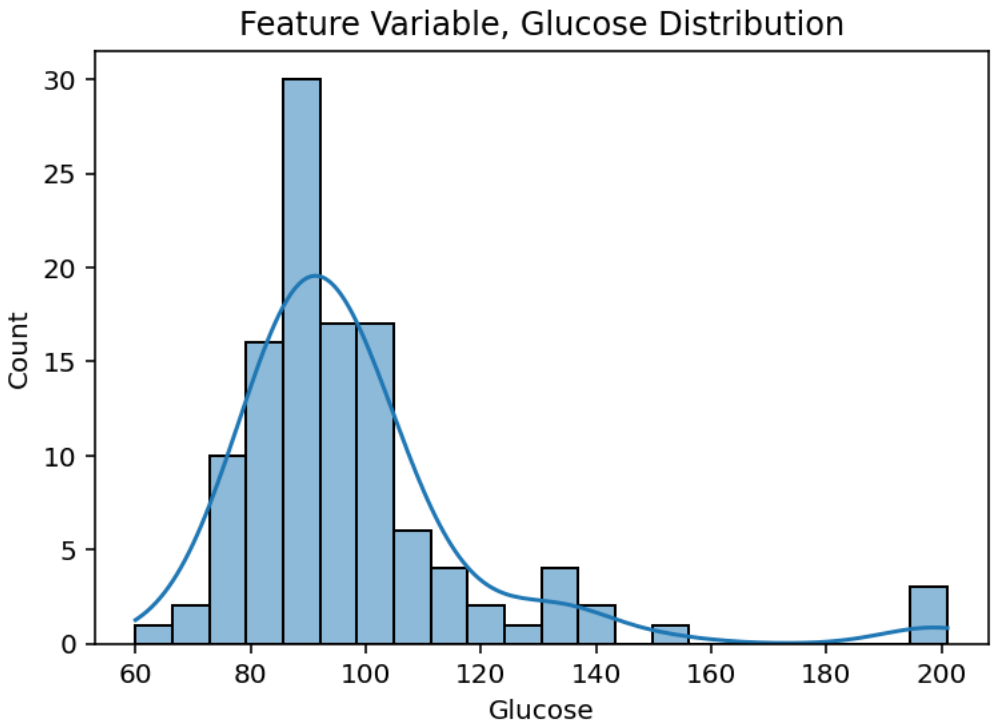
This model has the highest accuracy at: 0.8. It also produces the highest F1 score which is a balance between precision and recall.
Based on the top node in the decision tree, we can see the most important variable is glucose

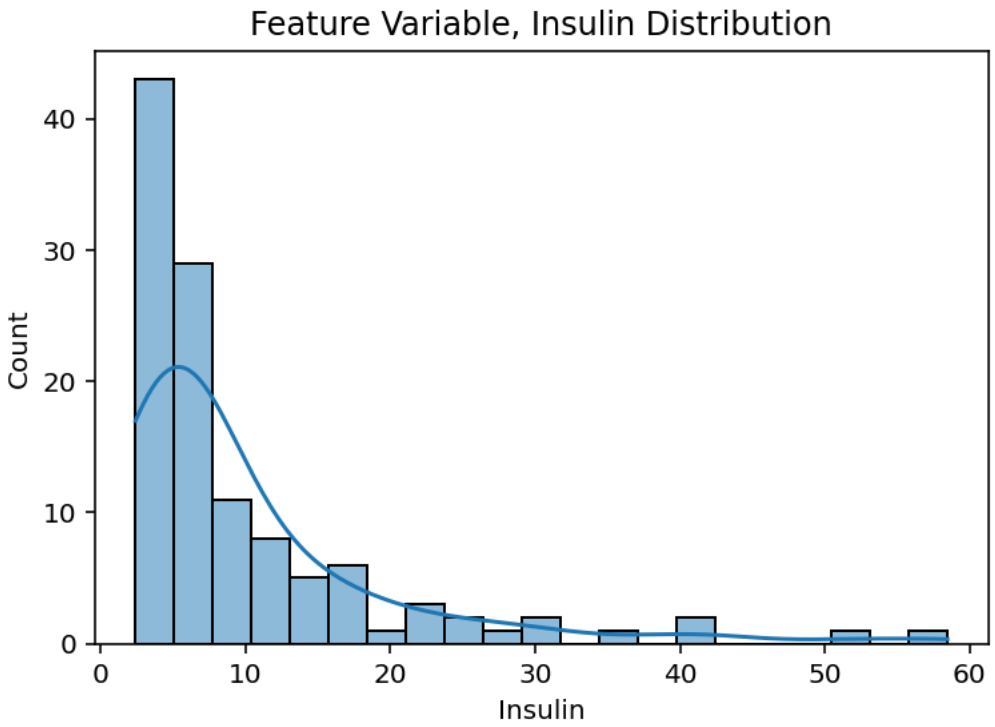
Relevant graphs:

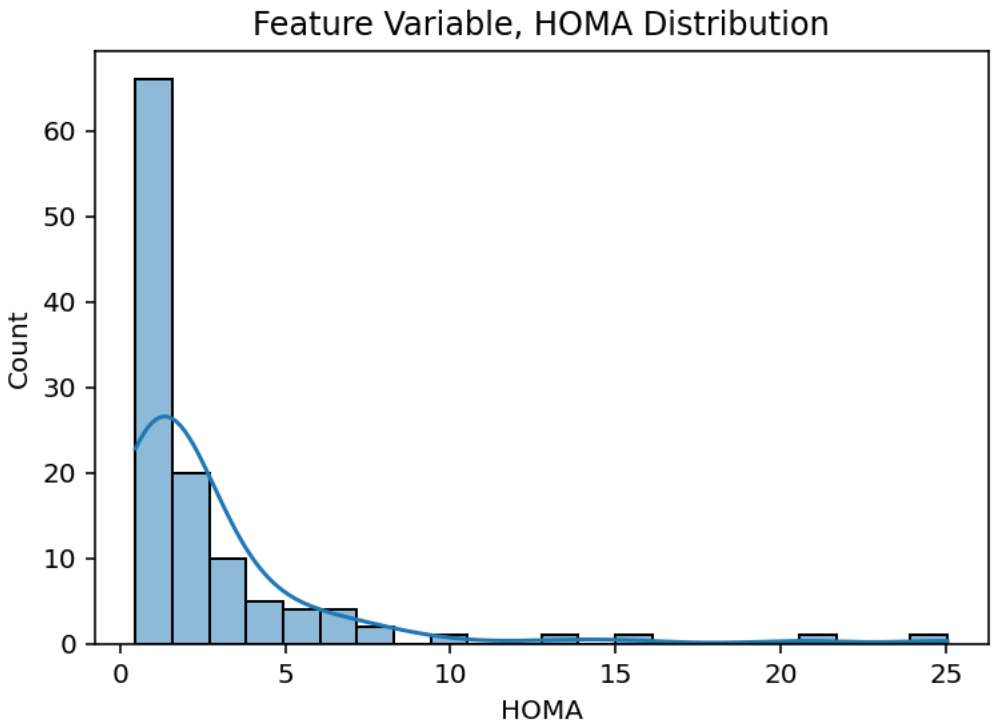


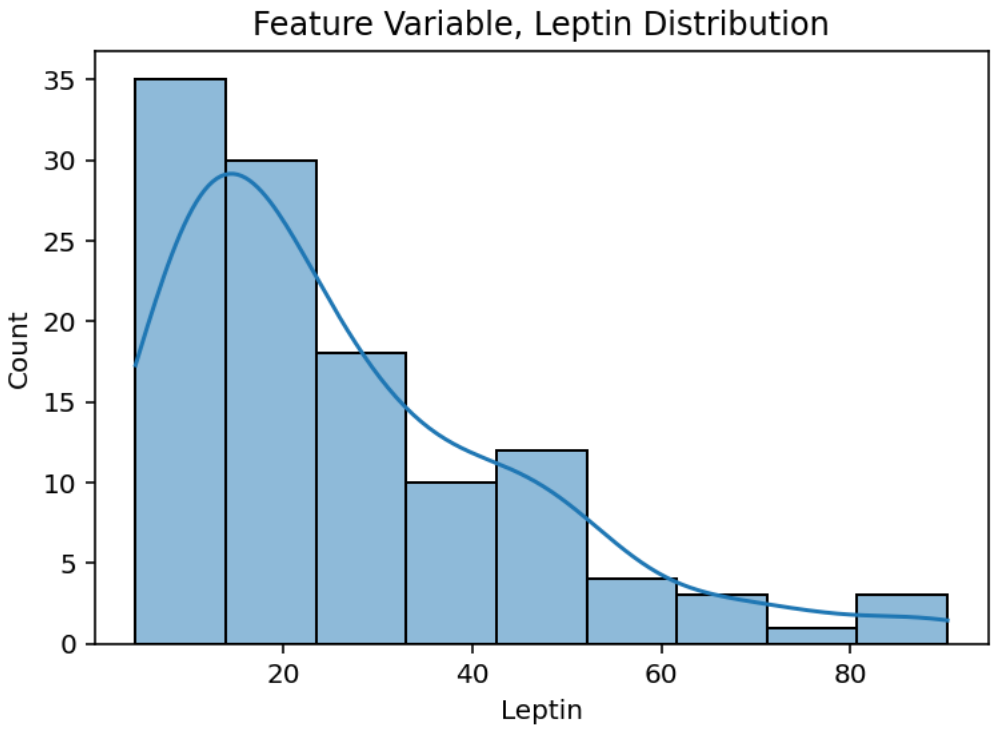

Feature Variable, Age Distribution
Count
Age
0
5
10
15
20
25
30
40
50
60
70
80
90

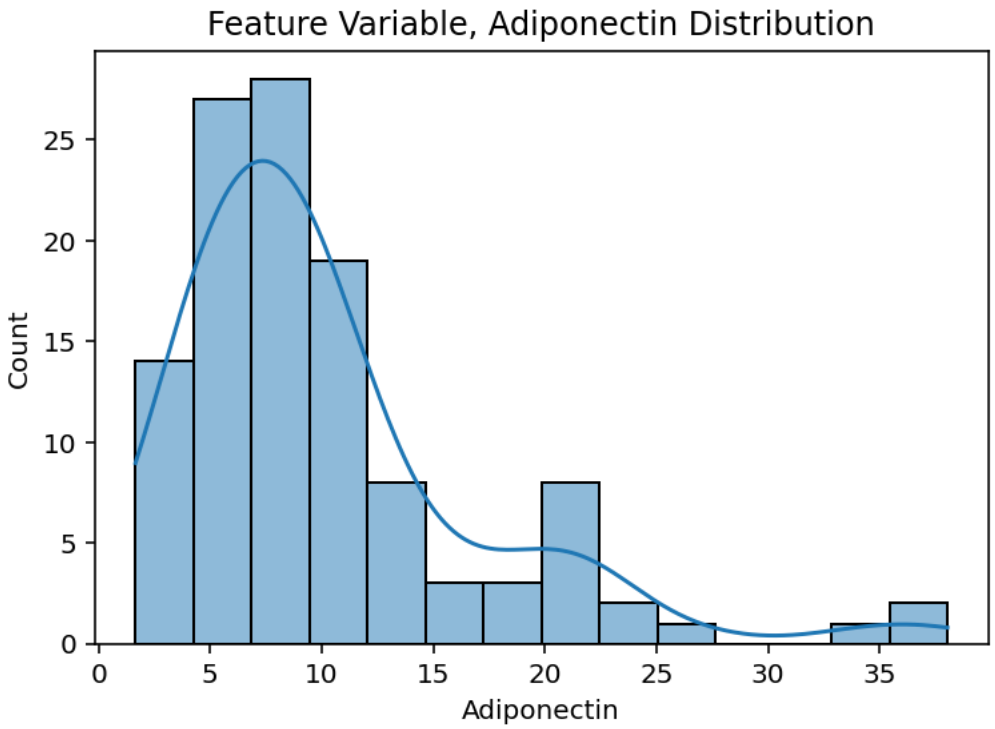

Feature Variable, BMI Distribution
Count
BMI
0
5
10
15
20
25
17.5
20.0
22.5
25.0
27.5
30.0
32.5
35.0
37.5

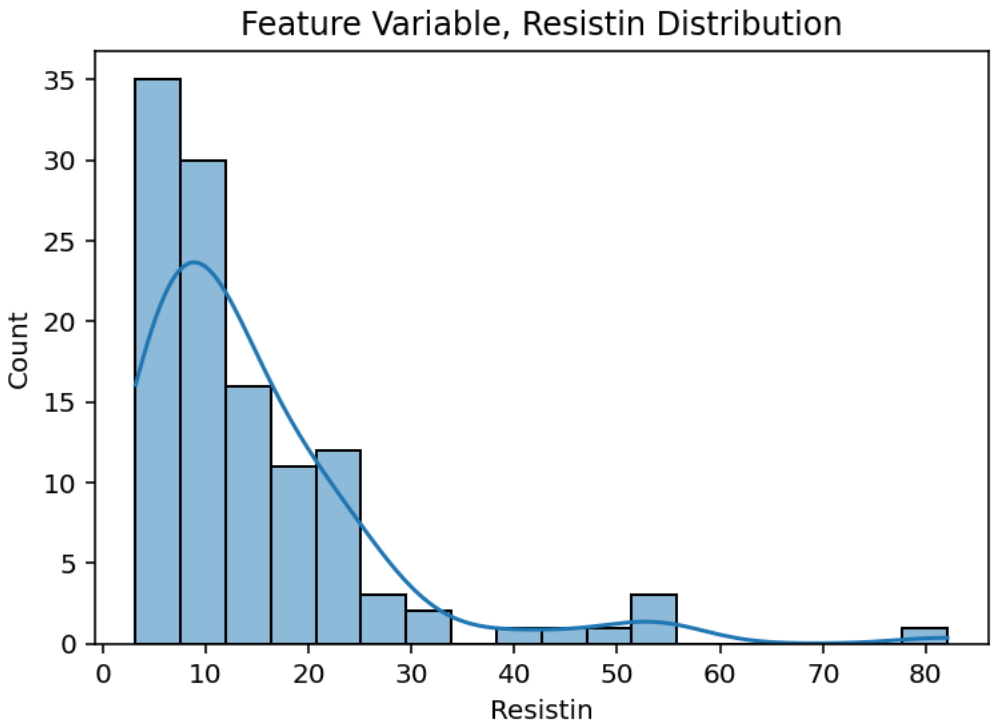
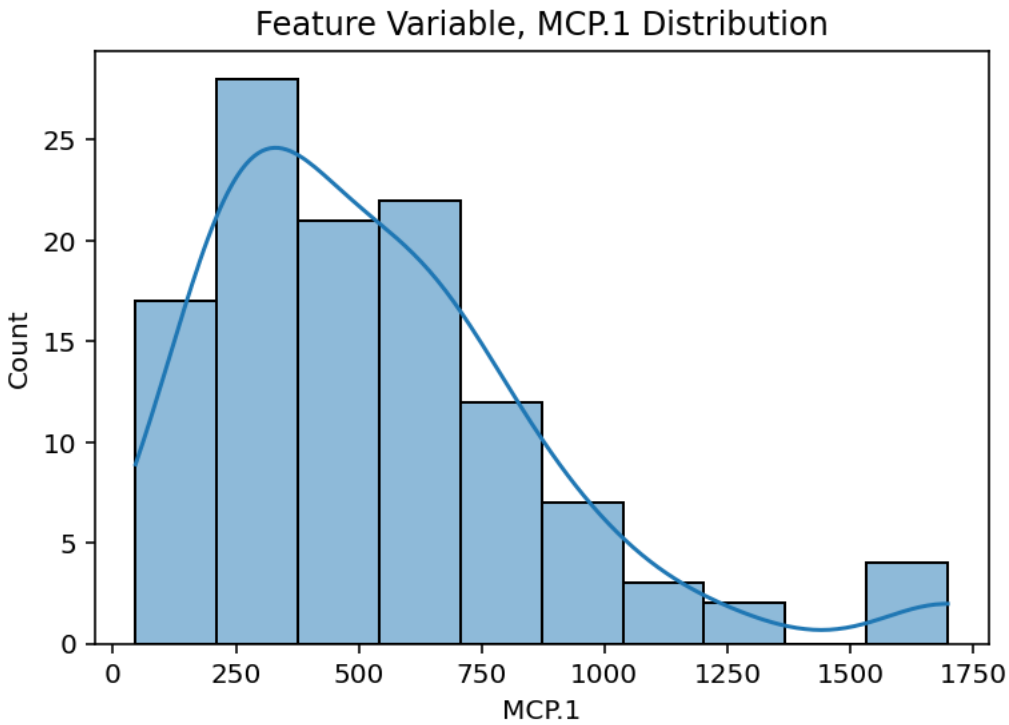
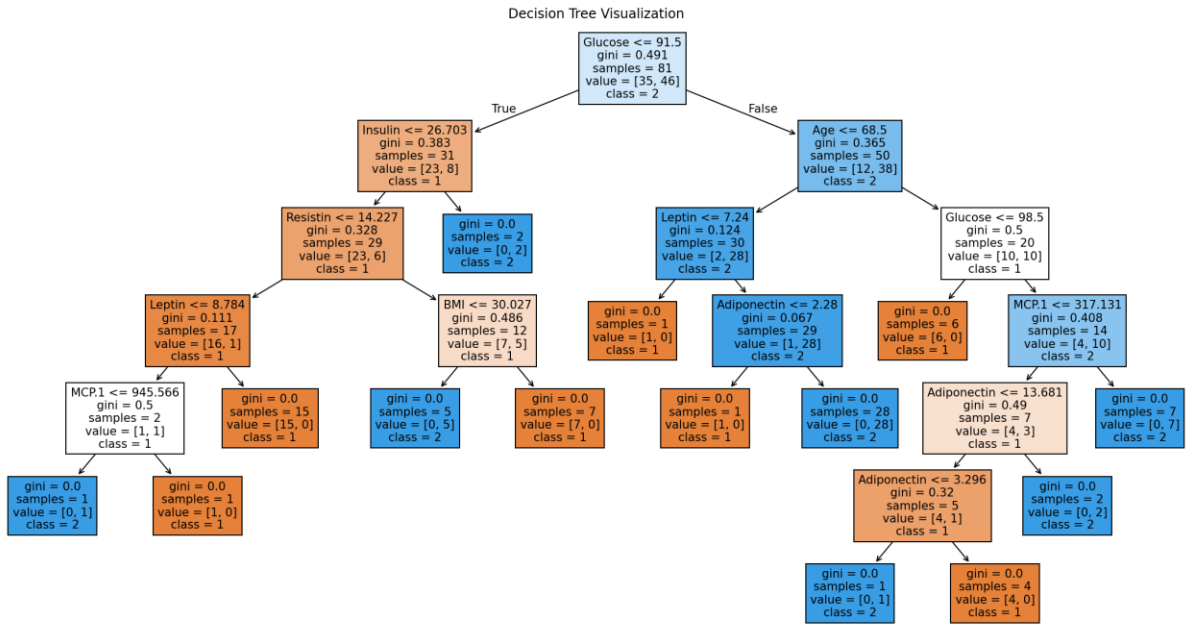

Feature Variable, Glucose Distribution
Count
Glucose
0
5
10
15
20
25
30
60
80
100
120
140
160
180
200


Feature Variable, Insulin Distribution
Count
Insulin
0
10
20
30
40
0
10
20
30
40
50
60


Feature Variable, HOMA Distribution
Count
HOMA
0
10
20
30
40
50
60
0
5
10
15
20
25


Feature Variable, Leptin Distribution
Count
Leptin
0
5
10
15
20
25
30
35
20
40
60
80


Feature Variable, Adiponectin Distribution
Count
Adiponectin
0
5
10
15
20
25
30
35


Feature Variable, Resistin Distribution
Count
Resistin
0
10
20
30
40
50
60
70
80

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